Econ 104 Project 1: Investigating Determinants of Wage

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```r
library(AER)
```

```
## Loading required package: car

## Loading required package: carData

## Loading required package: lmtest

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric

## Loading required package: sandwich

## Loading required package: survival
```

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following object is masked from 'package:car':
##
##     recode

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(car)
library(MASS)
```

```
##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##     select
```

```r
library(lmtest)
library(sandwich)
```

1: Introduction

This report investigates the main determinants of hourly wages in the 1985 US labor market using cross-sectional data from the Current Population Survey. Specifically, it examines how wages vary with worker characteristics such as education, work experience, gender, occupation, union status etc, in order to identify which factors are most strongly associated with earnings.

Understanding the drivers of wages is important for labor market policy and discussions surrounding inequality, as it helps distinguish whether differences in pay are primarily driven by human capital or broader structural factors. This distinction can help inform policy decisions related to education, labor protections, and wage inequality.

Prior to estimation, it is expected that wages will increase with education, work experience, union membership, and higher-paying occupations, showing the role of human capital and job characteristics in determining productivity and bargaining power. In contrast, variables such as marital status and ethnicity are expected to have a weaker association with wages.

2: Data Preparation and Train/Test Split

The dataset used is drawn from the Current Population Survey (CPS) 1985 Wage Data, available through the AER package (StatLib, https://lib.stat.cmu.edu/datasets/CPS_85_Wages ; Berndt, 1991). The dataset is loaded into R and inspected to determine the number of observations, variable types, and the presence of missing values.

```r
data("CPS1985", package = "AER")
cps <- CPS1985
str(cps)
```

```
## 'data.frame':    534 obs. of  11 variables:
##  $ wage      : num  5.1 4.95 6.67 4 7.5 ...
##  $ education : num  8 9 12 12 12 13 10 12 16 12 ...
##  $ experience: num  21 42 1 4 17 9 27 9 11 9 ...
##  $ age       : num  35 57 19 22 35 28 43 27 33 27 ...
##  $ ethnicity : Factor w/ 3 levels "cauc","hispanic",..: 2 1 1 1 1 1 1 1 1 1 ...
##  $ region    : Factor w/ 2 levels "south","other": 2 2 2 2 2 2 1 2 2 2 ...
##  $ gender    : Factor w/ 2 levels "male","female": 2 2 1 1 1 1 1 1 1 1 ...
```

```
##  $ occupation: Factor w/ 6 levels "worker","technical",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ sector    : Factor w/ 3 levels "manufacturing",..: 1 1 1 3 3 3 3 3 1 3 ...
##  $ union     : Factor w/ 2 levels "no","yes": 1 1 1 1 1 2 1 1 1 1 ...
##  $ married   : Factor w/ 2 levels "no","yes": 2 2 1 1 2 1 1 1 2 1 ...
```

```r
nrow(cps)
```

```
## [1] 534
```

```r
ncol(cps)
```

```
## [1] 11
```

```r
sum(is.na(cps))
```

```
## [1] 0
```

The dataset contains 534 observations and 11 variables. Variables are both numeric (eg. wage, education, experience) and categorical (eg. gender, occupation, union status).

There were no missing values detected across the variables used in the model, suggesting the data might have been precleaned. Because no missing values were present, listwise deletion was not required, avoiding potential bias or loss of information.

```r
var_table <- data.frame(
  Variable = c("wage", "education", "experience", "age", "ethnicity", "region", "gender",
               "union", "married", "occupation", "sector"),
  Type = c("Numeric", "Numeric", "Numeric", "Numeric", "Categorical", "Categorical",
           "Categorical", "Categorical", "Categorical", "Categorical", "Categorical"),
  Role = c("Response", "Predictor", "Predictor", "Predictor", "Predictor", "Predictor",
           "Predictor", "Predictor", "Predictor", "Predictor", "Predictor")
)
var_table
```

```
##      Variable        Type      Role
## 1        wage     Numeric  Response
## 2   education     Numeric Predictor
## 3  experience     Numeric Predictor
## 4         age     Numeric Predictor
## 5   ethnicity Categorical Predictor
## 6      region Categorical Predictor
## 7      gender Categorical Predictor
## 8       union Categorical Predictor
## 9     married Categorical Predictor
## 10 occupation Categorical Predictor
## 11     sector Categorical Predictor
```

The dataset is then divided into training and test sets using an 80/20 split. All model estimation and variable selection are conducted using the training set only.

```r
set.seed(123)

n <- nrow(cps)

train_index <- sample(seq_len(n), size = floor(0.8 * n))

train <- cps[train_index, ]
test  <- cps[-train_index, ]
```

The training set contains 427 observations, while the remaining 107 are reserved for the test set. This method reduces the risk of overfitting. However, a limitation for this specific dataset is that given the already relatively small sample size, the test set may not fully represent the population, which can introduce variability in performance metrics.

Categorical variables such as gender, union status, occupation, and sector were converted to factor variables, allowing R to generate dummy variables and interpret coefficients relative to a reference category. A potential limitation is that coefficient interpretations depend on the chosen reference category. In addition, categories with few observations may lead to less precise coefficient estimates.

```r
train$region     <- as.factor(train$region)
train$gender     <- as.factor(train$gender)
train$union      <- as.factor(train$union)
train$married    <- as.factor(train$married)
train$ethnicity  <- as.factor(train$ethnicity)
train$occupation <- as.factor(train$occupation)
train$sector     <- as.factor(train$sector)

test$region     <- as.factor(test$region)
test$gender     <- as.factor(test$gender)
test$union      <- as.factor(test$union)
test$married    <- as.factor(test$married)
test$ethnicity  <- as.factor(test$ethnicity)
test$occupation <- as.factor(test$occupation)
test$sector     <- as.factor(test$sector)
```

The dataset did not contain missing values, so no rows were dropped or imputation methods were required. Since the data appears to be pre-cleaned, no additional data cleaning steps were necessary.

3: Targeted Exploratory Analysis

With the data cleaned and split into training and test sets, the analysis proceeds to targeted exploratory analysis. This includes a five-number summary, selected diagnostic visualizations, and a correlation matrix to better understand the distribution of variables and relationships relevant to the research question.

The five-number summary provides an overview of the distribution of the numeric variables in the training dataset.

```r
summary(train[, c("wage", "education", "experience", "age")])
```

```
##       wage          education       experience         age
##  Min.   : 1.000   Min.   : 2.00   Min.   : 0.00   Min.   :18.00
##  1st Qu.: 5.250   1st Qu.:12.00   1st Qu.: 8.00   1st Qu.:28.00
```

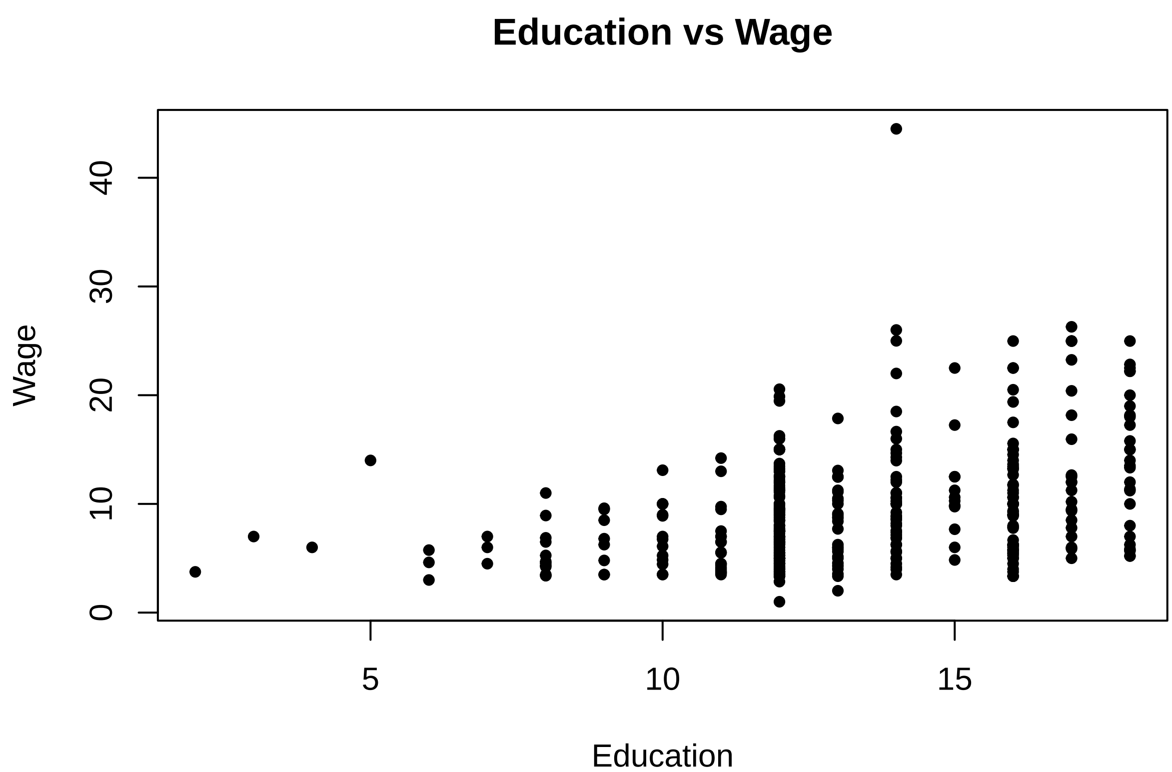
```
##  Median : 7.700   Median :12.00   Median :14.00   Median :34.00
##  Mean   : 9.069   Mean   :13.11   Mean   :17.68   Mean   :36.78
##  3rd Qu.:11.250   3rd Qu.:16.00   3rd Qu.:25.00   3rd Qu.:43.50
##  Max.   :44.500   Max.   :18.00   Max.   :55.00   Max.   :64.00
```

Hourly wages are right skewed, with a median of 7.70 dollars and a higher mean of 9.07 dollars, indicating that high income observations are pulling the mean above the median. Education is concentrated around 12–16 years, while experience and age show wide variation, suggesting a diverse workforce in terms of career stage.

```r
hist(train$wage,
     main = "Distribution of Wage",
     xlab = "Wage",
     col = c("#B3CFF5"))
```

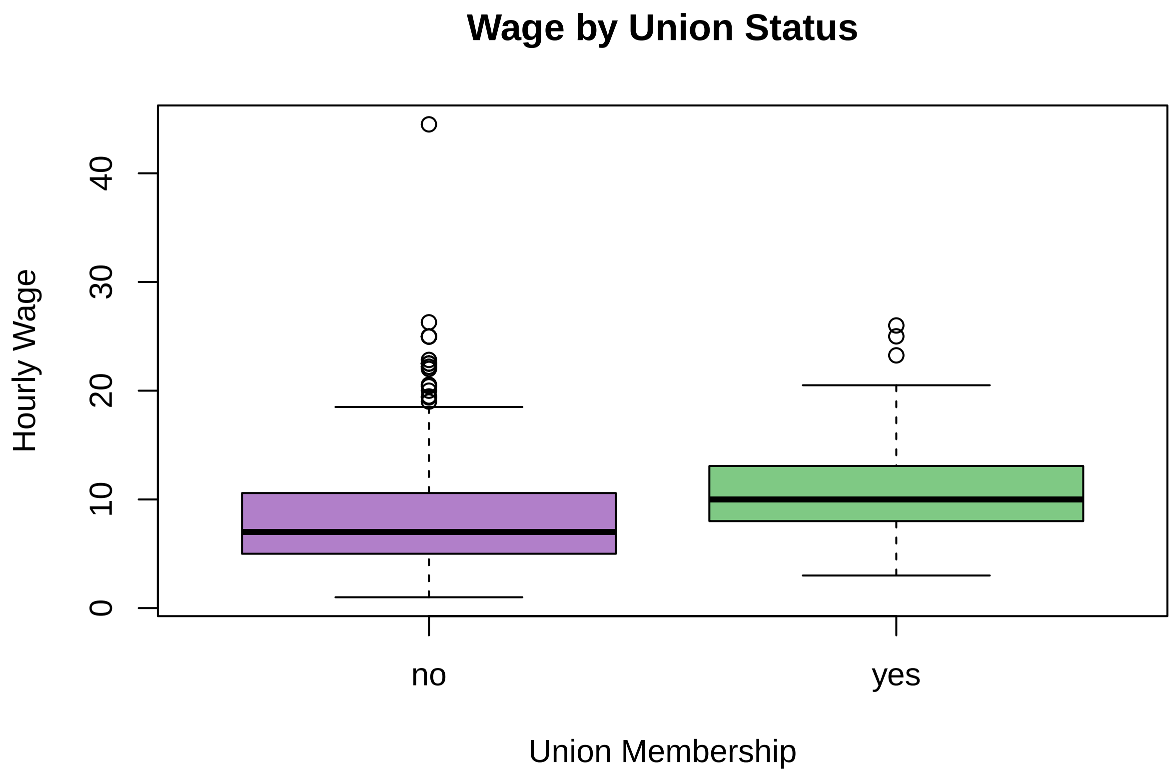
The wage distribution is right-skewed, indicating the presence of high-income outliers and suggesting potential heteroskedasticity, as the variance of wages appears larger at higher values

```r
plot(train$education, train$wage,
     main = "Education vs Wage",
     xlab = "Education",
     ylab = "Wage",
     pch = 20)
```



A scatterplot is appropriate for visualizing the relationship between education and wages, as both are numeric variables. However, since education takes on discrete values, the data appear in vertical clusters, which may obscure some variation due to overlapping points. Wages generally increase with education, indicating a positive relationship between years of schooling and earnings. However, the spread of wages becomes wider at higher education levels, suggesting greater variability in outcomes and possible heteroskedasticity.

```r
boxplot(wage ~ union, data = train,
        main = "Wage by Union Status",
        xlab = "Union Membership",
        ylab = "Hourly Wage",
        col = c("#B17FC9", "#7FC984"))
```



Workers who are union members have higher median wages than non-union workers, indicating a positive association between union membership and earnings. The distribution for union workers is also shifted upward overall, suggesting consistently higher wages across the group.

```
cor(train[, c("education", "experience", "age")])
```

```
##             education experience        age
## education   1.0000000 -0.3769676 -0.1691868
## experience -0.3769676  1.0000000  0.9765215
## age        -0.1691868  0.9765215  1.0000000
```

Experience and age are highly correlated ($|r| = 0.98 > 0.7$ threshold), indicating severe multicollinearity between these variables. This suggests that including both in the regression may lead to unstable coefficient estimates

Overall, the exploratory analysis indicates two key concerns for the regression model: potential heteroskedasticity and severe multicollinearity between experience and age. These observations will be formally tested in later steps.

4: Baseline Multiple Regression

A baseline OLS model is estimated using the training data.

```r
model_baseline <- lm(wage ~ education + experience + age + region + gender +
                       union + married + ethnicity + occupation + sector,
                     data = train)
summary(model_baseline)
```

```
##
## Call:
## lm(formula = wage ~ education + experience + age + region + gender +
##     union + married + ethnicity + occupation + sector, data = train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -11.191  -2.387  -0.714   1.908  34.827
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)            -0.23876    6.80475  -0.035 0.972027
## education               0.66817    1.11697   0.598 0.550042
## experience              0.11600    1.10951   0.105 0.916782
## age                    -0.04019    1.10834  -0.036 0.971090
## regionother             0.59719    0.47830   1.249 0.212537
## genderfemale           -1.68977    0.47597  -3.550 0.000430 ***
## unionyes                1.68548    0.57856   2.913 0.003772 **
## marriedyes              0.34072    0.46557   0.732 0.464689
## ethnicityhispanic      -0.41001    0.95274  -0.430 0.667166
## ethnicityother         -0.57426    0.68832  -0.834 0.404604
## occupationtechnical     2.40958    0.85495   2.818 0.005060 **
## occupationservices     -0.59012    0.77143  -0.765 0.444733
## occupationoffice        0.09868    0.77507   0.127 0.898754
## occupationsales        -0.95811    0.99492  -0.963 0.336115
## occupationmanagement    3.42366    0.90551   3.781 0.000179 ***
## sectorconstruction     -0.30801    1.14368  -0.269 0.787822
## sectorother            -1.04541    0.63672  -1.642 0.101384
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.38 on 410 degrees of freedom
## Multiple R-squared:  0.3215, Adjusted R-squared:  0.295
## F-statistic: 12.14 on 16 and 410 DF,  p-value: < 2.2e-16
```

Education is selected as the most economically important predictor, as it is a key measure of human capital. The coefficient on education is positive (0.668), meaning that an additional year of education is associated with a $0.67 increase in hourly wages, holding other factors constant. This is consistent with the hypothesis that higher human capital leads to higher earnings. However, the coefficient is not statistically significant. This is may be due to multicollinearity with experience and age, which inflates standard errors and makes it hard to isolate the individual effect of education.

The adjusted R squared is 0.295, meaning that the model explains approximately 29.5% of the variation in wages, indicating a moderate fit given the cross-sectional nature of the data. The F-statistic is 12.14 (p-value <2.2e-16), suggesting that the model is jointly significant and that at least some predictors are important in explaining wage variation.

It is surprising that education, experience, and age are not statistically significant, as these are typically strong predictors of wages. This is again likely due to high multicollinearity among these variables. In contrast, gender and union status are statistically significant, with females earning less and union members earning more, which aligns with expectations regarding labor market disparities and bargaining power.

5: Multicollinearity — VIF Screen

A VIF threshold of 5 is used to detect multicollinearity, as it captures moderate collinearity without being overly permissive. This is preferred over a threshold of 10, which may miss levels of collinearity that still inflate standard errors and distort inference. Since the goal is to identify the strongest determinants of wages, each predictor must provide reliable and independent information. The adjusted measure GVIF^(1/(2 x Df)) is used for interpretation, as it places all variables on a comparable scale regardless of their number of degrees of freedom.

```
vif(model_baseline)
```

```
##                   GVIF Df GVIF^(1/(2*Df))
## education   200.825077  1       14.171276
## experience 4145.887733  1       64.388568
## age        3661.552924  1       60.510767
## region        1.058954  1        1.029055
## gender        1.245075  1        1.115829
## union         1.112124  1        1.054573
## married       1.102570  1        1.050033
## ethnicity     1.109190  2        1.026246
## occupation    3.160634  5        1.121960
## sector        1.479519  2        1.102885
```

In the baseline model, experience, age, and education all exceed the threshold, indicating severe multicollinearity. Variables are evaluated sequentially, beginning with those exhibiting the highest VIF. Although experience had the highest VIF (64.39), we made the decision to remove age instead. This is because experience provides a more direct and economically meaningful measure of labor market exposure, while age is broader.

```
model_vif <- lm(wage ~ education + experience + region + gender + union +
                  married + ethnicity + occupation + sector,
                data = train)

vif(model_vif)
```

```
##                GVIF Df GVIF^(1/(2*Df))
## education  2.136525  1        1.461686
## experience 1.401789  1        1.183972
## region     1.058398  1        1.028785
## gender     1.241033  1        1.114017
## union      1.111942  1        1.054487
## married    1.099124  1        1.048391
## ethnicity  1.108817  2        1.026160
## occupation 3.132593  5        1.120961
## sector     1.479297  2        1.102843
```

After removing age, all remaining predictors fell below the threshold, with adjusted GVIF values close to 1. This indicates that multicollinearity was resolved.

```r
summary(model_vif)
```

```
##
## Call:
## lm(formula = wage ~ education + experience + region + gender +
##     union + married + ethnicity + occupation + sector, data = train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -11.191  -2.387  -0.715   1.908  34.825
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -0.47806    1.65981  -0.288 0.773478
## education             0.62788    0.11507   5.457 8.41e-08 ***
## experience            0.07577    0.02038   3.719 0.000228 ***
## regionother           0.59758    0.47759   1.251 0.211557
## genderfemale         -1.68879    0.47461  -3.558 0.000417 ***
## unionyes              1.68521    0.57781   2.917 0.003733 **
## marriedyes            0.33978    0.46428   0.732 0.464685
## ethnicityhispanic    -0.41064    0.95143  -0.432 0.666255
## ethnicityother       -0.57423    0.68748  -0.835 0.404054
## occupationtechnical   2.41153    0.85221   2.830 0.004887 **
## occupationservices   -0.59046    0.77043  -0.766 0.443881
## occupationoffice      0.09816    0.77400   0.127 0.899144
## occupationsales      -0.95814    0.99371  -0.964 0.335511
## occupationmanagement  3.42379    0.90441   3.786 0.000176 ***
## sectorconstruction   -0.30753    1.14222  -0.269 0.787879
## sectorother          -1.04524    0.63593  -1.644 0.101015
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.375 on 411 degrees of freedom
## Multiple R-squared:  0.3215, Adjusted R-squared:  0.2967
## F-statistic: 12.98 on 15 and 411 DF,  p-value: < 2.2e-16
```

The coefficients on education and experience remain similar in magnitude compared to the baseline model, but their standard errors decrease substantially, making both variables statistically significant. This suggests that their lack of significance in the baseline model was mainly driven by multicollinearity.

The result is not too surprising because, as shown in the CPS 1985 dataset, potential experience is not directly observed, but is calculated as (age- education- 6). This means experience is a linear combination of age and education. As a result, including all three variables in the model creates severe multicollinearity, making it difficult for OLS to distinguish whether a wage change is driven by someone being older, more experienced, or more educated, as they all move together. Once age is dropped, that dependency disappears, allowing experience and education to be interpreted independently. Experience can now capture the effect of time spent in the labor market on wages, while education reflects human capital. In addition, the remaining predictors, including gender, occupation, and union status, were unaffected by the removal, which is consistent with their low baseline VIF values and confirms that the multicollinearity problem was isolated entirely to the relationship between experience, age, and education.

6: Model Selection — AIC

We used stepwise selection using the stepAIC() function, starting from the baseline model.

```r
stepAIC(model_baseline)
```

```
## Start:  AIC=1278.13
## wage ~ education + experience + age + region + gender + union +
##     married + ethnicity + occupation + sector
##
##              Df Sum of Sq    RSS    AIC
## - ethnicity   2     15.77 7882.8 1275.0
## - age         1      0.03 7867.1 1276.1
## - experience  1      0.21 7867.3 1276.1
## - education   1      6.87 7873.9 1276.5
## - married     1     10.28 7877.4 1276.7
## - sector      2     53.15 7920.2 1277.0
## - region      1     29.91 7897.0 1277.8
## <none>                    7867.1 1278.1
## - union       1    162.85 8029.9 1284.9
## - gender      1    241.84 8108.9 1289.1
## - occupation  5    575.78 8442.9 1298.3
##
## Step:  AIC=1274.99
## wage ~ education + experience + age + region + gender + union +
##     married + occupation + sector
##
##              Df Sum of Sq    RSS    AIC
## - age         1      0.03 7882.9 1273.0
## - experience  1      0.23 7883.1 1273.0
## - education   1      7.17 7890.0 1273.4
## - married     1     10.63 7893.5 1273.6
## - sector      2     50.88 7933.7 1273.7
## - region      1     34.29 7917.1 1274.8
## <none>                    7882.8 1275.0
## - union       1    155.46 8038.3 1281.3
## - gender      1    236.00 8118.8 1285.6
## - occupation  5    583.69 8466.5 1295.5
##
## Step:  AIC=1272.99
## wage ~ education + experience + region + gender + union + married +
##     occupation + sector
##
##              Df Sum of Sq    RSS    AIC
## - married     1     10.60 7893.5 1271.6
## - sector      2     50.87 7933.7 1271.7
## - region      1     34.36 7917.2 1272.8
## <none>                    7882.9 1273.0
## - union       1    155.42 8038.3 1279.3
## - gender      1    236.46 8119.3 1283.6
## - experience  1    270.48 8153.4 1285.4
## - occupation  5    585.61 8468.5 1293.6
## - education   1    612.52 8495.4 1302.9
```

```
##
## Step:  AIC=1271.56
## wage ~ education + experience + region + gender + union + occupation +
##     sector
##
##              Df Sum of Sq    RSS    AIC
## - sector      2     52.35 7945.8 1270.4
## - region      1     32.16 7925.6 1271.3
## <none>                    7893.5 1271.6
## - union       1    159.12 8052.6 1278.1
## - gender      1    236.38 8129.9 1282.2
## - experience  1    314.43 8207.9 1286.2
## - occupation  5    594.10 8487.6 1292.5
## - education   1    616.52 8510.0 1301.7
##
## Step:  AIC=1270.39
## wage ~ education + experience + region + gender + union + occupation
##
##              Df Sum of Sq    RSS    AIC
## - region      1     33.87 7979.7 1270.2
## <none>                    7945.8 1270.4
## - union       1    158.77 8104.6 1276.8
## - gender      1    227.22 8173.0 1280.4
## - experience  1    369.18 8315.0 1287.8
## - occupation  5    597.04 8542.9 1291.3
## - education   1    635.55 8581.4 1301.2
##
## Step:  AIC=1270.2
## wage ~ education + experience + gender + union + occupation
##
##              Df Sum of Sq    RSS    AIC
## <none>                    7979.7 1270.2
## - union       1    172.42 8152.1 1277.3
## - gender      1    224.14 8203.8 1280.0
## - experience  1    383.62 8363.3 1288.2
## - occupation  5    609.99 8589.7 1291.7
## - education   1    673.12 8652.8 1302.8
```

```
##
## Call:
## lm(formula = wage ~ education + experience + gender + union +
##     occupation, data = train)
##
## Coefficients:
##          (Intercept)             education            experience
##             -1.05857               0.66110               0.08666
##         genderfemale              unionyes   occupationtechnical
##             -1.61041               1.72005               1.98382
##   occupationservices      occupationoffice       occupationsales
##             -1.21173              -0.47545              -1.36162
## occupationmanagement
##              2.97266
```

The AIC-selected model includes education, experience, gender, union, and occupation. The model dropped age, region, ethnicity, sector, and married. Similar to the VIF model, age is dropped as it becomes largely redundant with experience already capturing time spent in the labor market. Moreover, married, region, ethnicity, and sector were also removed, suggesting that they contributed little to wage variation.

```r
model_AIC <-lm(wage ~ education + experience + gender + union + occupation ,
               data = train)
summary(model_AIC)
```

```
##
## Call:
## lm(formula = wage ~ education + experience + gender + union +
##     occupation, data = train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -10.927  -2.492  -0.702   1.946  34.854
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)           -1.05857    1.51102  -0.701 0.483964
## education              0.66110    0.11147   5.931 6.33e-09 ***
## experience             0.08666    0.01935   4.477 9.77e-06 ***
## genderfemale          -1.61041    0.47054  -3.422 0.000682 ***
## unionyes               1.72005    0.57302   3.002 0.002846 **
## occupationtechnical    1.98382    0.78915   2.514 0.012317 *
## occupationservices    -1.21173    0.68481  -1.769 0.077551 .
## occupationoffice      -0.47545    0.70997  -0.670 0.503440
## occupationsales       -1.36162    0.95297  -1.429 0.153805
## occupationmanagement  2.97266    0.85321   3.484 0.000546 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.374 on 417 degrees of freedom
## Multiple R-squared:  0.3118, Adjusted R-squared:  0.2969
## F-statistic: 20.99 on 9 and 417 DF,  p-value: < 2.2e-16
```

The AIC-selected model partially agrees with our VIF-trimmed model from step 5, as both dropped age and retained experience, which matches our earlier decision that experience is the more theoretically grounded variable in a wage regression. However, AIC went further by dropping region, ethnicity, sector, and marital status, which were retained in the VIF-trimmed model. This is because AIC optimizes predictive fit and removes variables that are statistically redundant rather than addressing multicollinearity, so they keep and drop different variables.

```r
comparison_table <- data.frame(
  Model = c("Baseline", "VIF_trimmed", "AIC_selected"),

  AIC = c(AIC(model_baseline), AIC(model_vif), AIC(model_AIC)),

  BIC = c(BIC(model_baseline), BIC(model_vif), BIC(model_AIC)),

  Adj_R2 = c(summary(model_baseline)$adj.r.squared,
             summary(model_vif)$adj.r.squared,
             summary(model_AIC)$adj.r.squared)
```

```
)

comparison_table
```

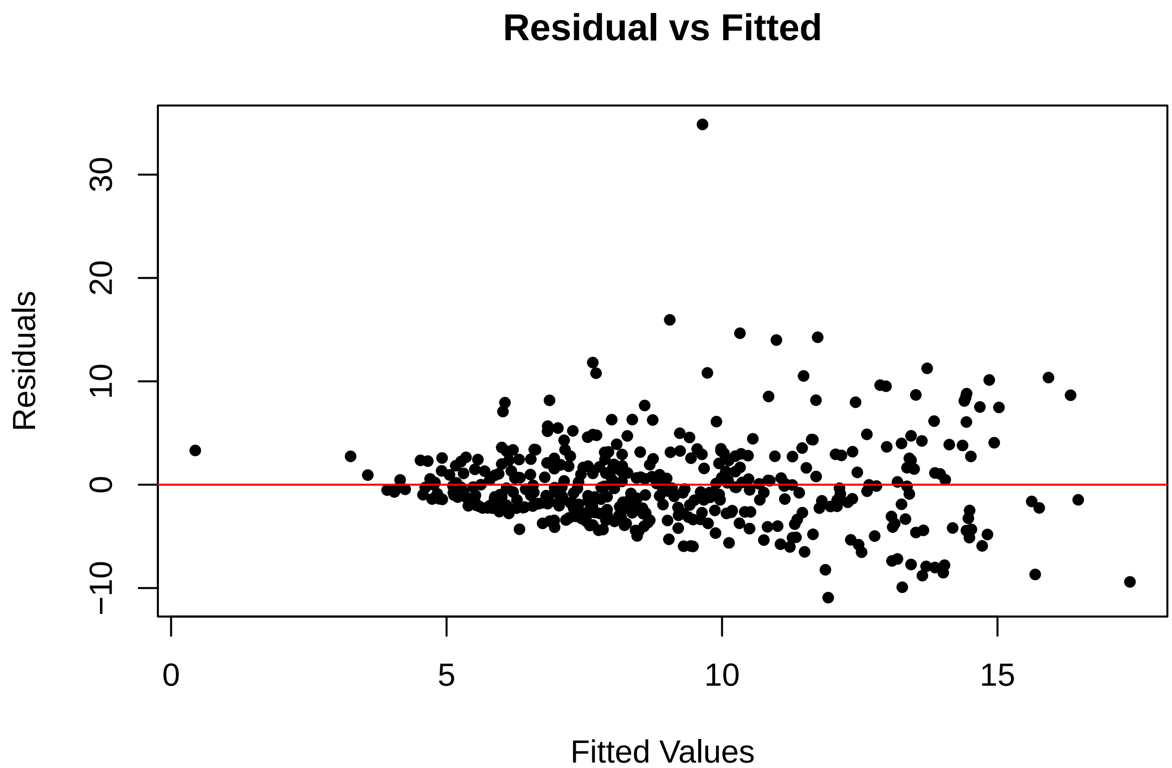
```
##           Model      AIC      BIC    Adj_R2
## 1      Baseline 2491.905 2564.927 0.2950299
## 2   VIF_trimmed 2489.907 2558.872 0.2967429
## 3  AIC_selected 2483.975 2528.599 0.2969415
```

Comparing the models, the AIC-selected model has the lowest AIC and BIC, indicating that it might be the best trade-off between the model fit and complexity among the candidates. Moreover, it also has a slightly lower R^2, which shows a marginal improvement in explanatory power. Lastly, the AIC-selected model is simpler with fewer predictors, which is easier to interpret compared to the VIF-trimmed model. Therefore, the AIC-selected model is preferred.

7:Residual Diagnostics

The residual versus fitted values plot shows signs of unequal variance. The majority of observations are clustered in the lower fitted value range from 0 to 10, with the spread of points becoming wider as fitted values increase, which creates a fan shape.

```r
plot(model_AIC$fitted.values, resid(model_AIC),
     main = "Residual vs Fitted",
     xlab = "Fitted Values",
     ylab = "Residuals",
     pch = 20)
abline(h=0, col = "#FF0000")
```



This fan shape is evidence of heteroskedasticity; the spread is narrow on the left and wide on the right, indicating that the variance of the residuals is not constant. The dots are tightly packed close to the zero line at the low fitted values between 0 and 5 with a small spread. For the high fitted values between 10 and 15, the spread increases, with residuals reaching as high as 35 and as low as -10.

There is no clear curvature in the plot, suggesting that non-linearity is not a major concern. However, there is one influential observation that stood out at the top of the plot, notably far above the rest of the residuals, located at a residual value of approximately 35. This point is a potential outlier that may affect the regression estimates.

```r
resettest(model_AIC, power = 2)
```

```
##
##  RESET test
##
## data:  model_AIC
## RESET = 0.14911, df1 = 1, df2 = 416, p-value = 0.6996
```

The RESET test yields a p-value of 0.6996, which is above the significance level. Therefore, we fail to reject the null hypothesis that the model is correctly specified. This shows that the functional form isn't misspecified, and the linear model appears appropriate.

8: Heteroskedasticity: Test and Correct

The residuals-versus-fitted plot suggests that the spread of the residuals increases as the fitted values rise, which is consistent with heteroskedasticity. Because this pattern appears to be a general increase in variance rather than a clear break between groups, the Breusch-Pagan (BP) test was used to formally test for heteroskedasticity.

```r
bptest(model_AIC)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  model_AIC
## BP = 23.319, df = 9, p-value = 0.005519
```

The BP test generates a p value of 0.0055, meaning that we reject the null hypothesis of constant variance. This provides statistical evidence for the presence of heteroskedasticity beyond what was visually observed on the residual vs fitted graph.

To address this, robust standard errors is used instead of applying a log transformation, as the RESET test did not indicate functional form misspecification. Robust standard errors can correct for heteroskedasticity without changing the model specification, making them more appropriate for this case.

```r
robust_results <-coeftest(model_AIC, vcov = vcovHC(model_AIC, type = "HC1"))
print (robust_results)
```

```
##
## t test of coefficients:
##
##                        Estimate Std. Error t value  Pr(>|t|)
## (Intercept)           -1.058572   1.567922 -0.6751 0.4999586
## education              0.661097   0.116505  5.6744 2.608e-08 ***
## experience             0.086660   0.023393  3.7045 0.0002403 ***
## genderfemale          -1.610407   0.519006 -3.1029 0.0020470 **
## unionyes               1.720054   0.594604  2.8928 0.0040187 **
## occupationtechnical    1.983817   0.805129  2.4640 0.0141432 *
## occupationservices    -1.211728   0.588008 -2.0607 0.0399479 *
## occupationoffice      -0.475447   0.549971 -0.8645 0.3878139
## occupationsales       -1.361620   0.676642 -2.0123 0.0448289 *
## occupationmanagement   2.972659   1.286284  2.3110 0.0213173 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(model_AIC)
```

```
##
## Call:
## lm(formula = wage ~ education + experience + gender + union +
##     occupation, data = train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -10.927  -2.492  -0.702   1.946  34.854
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)          -1.05857    1.51102  -0.701 0.483964
## education             0.66110    0.11147   5.931 6.33e-09 ***
## experience            0.08666    0.01935   4.477 9.77e-06 ***
## genderfemale         -1.61041    0.47054  -3.422 0.000682 ***
## unionyes              1.72005    0.57302   3.002 0.002846 **
## occupationtechnical   1.98382    0.78915   2.514 0.012317 *
## occupationservices   -1.21173    0.68481  -1.769 0.077551 .
## occupationoffice     -0.47545    0.70997  -0.670 0.503440
## occupationsales      -1.36162    0.95297  -1.429 0.153805
## occupationmanagement  2.97266    0.85321   3.484 0.000546 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.374 on 417 degrees of freedom
## Multiple R-squared:  0.3118, Adjusted R-squared:  0.2969
## F-statistic: 20.99 on 9 and 417 DF,  p-value: < 2.2e-16
```

After the correction, the significance of the coefficients did not change, but noticed an increase in the standard errors of all the coefficients. Therefore our inferences should be more conservative than if homoskedasticity held.

9: Final Model

While economic theory suggests that wages may exhibit diminishing returns to experience, which could be better modeled using a quadratic term, the RESET test does not indicate any functional form misspecification. Therefore, no additional nonlinear terms are included, and the Step 6 model is retained.

Final model: AIC-selected model

10: Out of Sample Evaluation

```r
test_prediction <- predict(model_AIC, newdata = test)

mae <- mean(abs(test$wage - test_prediction))
mse <- mean((test$wage - test_prediction)^2)

train_prediction <- predict(model_AIC, newdata = train)

in_sample_mae <- mean(abs(train$wage - train_prediction))
in_sample_mse <- mean((train$wage - train_prediction)^2)

comparison <- data.frame(
  Metric = c("MSE","MAE"),
  In_Sample = c(in_sample_mse, in_sample_mae),
  Out_of_Sample = c(mse, mae)
)
comparison
```

```
##   Metric In_Sample Out_of_Sample
## 1    MSE 18.687812     15.841937
## 2    MAE  3.054811      3.039044
```

In-sample MAE and MSE are 3.055 and 18.688, while test-set MAE and MSE are 3.039 and 15.842.

The out-of-sample errors are very similar to the in-sample errors, and in this case slightly smaller. This suggests that the model generalizes well to new data and does not exhibit signs of overfitting. The small difference between training and test errors is likely due to sampling variation instead of an improvement in predictive performance.

11: Executive Summary

This report analyzes the key determinants of hourly wages in the 1985 US labor market using cross-sectional data from the Current Population Survey. The main goal is to identify which worker characteristics are most strongly associated with earnings. Using regression analysis with AIC-based model selection, the results show that education, experience, gender, union membership, and occupation are the most important predictors of wages, consistent with the initial hypothesis. In particular, higher education and greater labor market experience are strongly associated with higher wages, while union membership and certain occupations also contribute positively.

These results suggest that governments should prioritise policies aimed at improving access to education and expanding opportunities for workforce experience, such as job training programs, internships, or apprenticeship schemes. These policies may be effective in increasing overall wages and supporting economic growth in the US.

However, the model also reveals heteroskedasticity, meaning that wage variability increases at higher income levels, requiring more cautious interpretation of statistical significance. Additionally, the analysis is limited by the use of cross-sectional data, which restricts the ability to make causal claims about the relationships observed. Other limitations include omitted variable bias, as unobserved factors such as individual ability or firm characteristics are not included and may influence wages.

AI Usage log

Step 3: Excluding categorical variables in 5 number summary I originally used summary(train) to generate the 5 number summary but it included the categorical variables as well. I asked Chat GPT to figure out how to exclude the categorical variables in 5 number summary.

Step 6: Comparison table I didn't know how to extract the AIC, BIC, adjusted R squared values for every model and make it into a comparison table. I originally turned to online R resources but many required additional packages or were too complicated. ChatGPT was used to provide guidance on what command would be most suitable for my needs.

AI Reflection: As I mainly used AI to help figure out how to code specific things, the AI would often be correct after I described the problem to them. However, if I submitted simple prompts like "How do I make a comparison table in R," the code it gave me would not be suitable for my needs for this project. I had to be a lot more specific for it to be helpful. For the 5 number summary case, AI gave me an overly long code suggestion on how to exclude the categorical variables. While it might not necessarily be wrong, I didn't like how AI's suggestions were often overly long and overly complicated, so I would turn to online sources instead and only used AI as a very last resort.

I also asked ChatGPT to critique my AIC-selected model compared to the baseline. It responded that the AIC model is "preferable because it achieves similar explanatory power with fewer variables, making it more efficient and easier to interpret while retaining the key economic drivers of wages". I agreed with this judgement, however it didn't bring up a concern I had about dropping variables may sacrifice some control for omitted factors, so it is better suited for prediction than for fully explaining wage determination.